ODOO MAKE-A-TON HACKATHON

OFFICIAL PROJECT BLUEPRINT

SUBSCRIPTION MANAGEMENT SYSTEM

FitFlow

Gym Membership & Subscription Management Platform

> A comprehensive, responsive operations dashboard and subscription engine built around Odoo's architectural principles. FitFlow runs the full lifecycle of a gym subscription business: product plans with dynamic variants, quotation templates, automated billing runs, real-time desk attendance check-in, 7-day expiry renewal alerts, and sub-2-second executive finance analytics.

🎯 The Hackathon Challenge

Direct implementation of the Brightloop Subscription Management challenge: plans, recurring billing intervals, variant pricing (+560 model), discount rules, and overdue invoice tracking.

Operational Reality

Eliminates paper binders, disconnected POS machines, and manual spreadsheets by creating a unified workflow from member onboarding to daily front-desk attendance.

📱 Responsive Ergonomics

Desktop command center for admins with keyboard shortcuts and two-column workflows, paired with a touch-friendly mobile bottom-nav view for quick front-desk check-in.

🔄 Odoo Native Alignment

Directly models Odoo's `sale.subscription`, `account.move`, `res.partner`, and `hr.attendance` workflows with modular business logic.

Event: Odoo Make-A-Ton | **Domain:** Fitness & Subscription Operations

Status: Architecture & Execution Blueprint | **Target Response:** < 2s

1. The Pitch Strategy & Hackathon Narrative

> *"Every gym in the world is fundamentally a subscription business fighting the exact same daily battles as SaaS platforms: failed recurring payments, expiring contracts, manual invoice typing, and customer churn. Today, gym staff juggle spreadsheets, WhatsApp reminders, and paper check-in sheets. FitFlow unites these operations into an Odoo-driven platform where quoting, recurring billing, desk check-ins, and renewals flow seamlessly."*

The 3-Minute Live Hackathon Demo Arc

Judges at Odoo Make-A-Ton look for business coherence, architectural elegance, and complete end-to-end execution. Here is the timed live demonstration sequence:

Timing	Demo Stage	What the Judges See On Screen	Odoo Challenge Criteria
0:00 – 0:45	**Admin Catalog & Setup**	Admin defines the "Elite Fitness" plan (Monthly/Quarterly/Yearly), creates a Personal Trainer variant (+₹500 / +$50), and configures a 10% promo discount code with tax calculation rules.	Plan variants, billing frequencies, pricing rules, tax rules.
0:45 – 1:30	**Quote to Confirmed Subscription**	Front desk staff creates a quotation from a pre-built template for a new member, applies the promo discount, and confirms it. System auto-generates the invoice and records payment. Status moves to **ACTIVE**.	Quotation lifecycle: Draft → Quote → Confirmed → Active. Instant invoice generation.
1:30 – 2:00	**Desk Operations & Attendance**	Staff switches to Mobile View. Taps "⚡ Quick Check-in", searches member, gets an instant green "Active" badge with validity date, and logs attendance. Demonstrates rejection alert for expired members.	Mobile responsive UI, touch-friendly front desk operations, real-time check-in validation.
2:00 – 2:30	**Billing Day & Renewal Engine**	Simulates "Billing Day": System flags memberships expiring within 7 days. One-click batch renewal creates the next cycle invoice and extends valid membership dates without manual re-typing.	Recurring billing automation, renewal alert engine, no duplicate billing.
2:30 – 3:00	**Finance Lead Executive Dashboard**	Admin reviews real-time KPIs: Active Subscriptions, Monthly Recurring Revenue (MRR), Overdue Invoices list with chasing notes, and inactive member retention alerts (>14 days absent).	Executive finance summary, overdue debt chasing, sub-2s responsiveness.

2. FitFlow & Odoo Ecosystem Mapping

FitFlow is built to seamlessly mirror the modular data structures and transaction lifecycles of Odoo ERP. The table below details how every gym operation directly maps to Odoo applications:

FitFlow Feature	Target Odoo App	Odoo Technical Model	Workflow & Architectural Role
Member Records & Enquiries	Odoo CRM & Contacts	`res.partner`, `crm.lead`	Central contact master, stage-based lead conversion, emergency details.
Membership Plans & Add-ons	Odoo Sales & Products	`product.template`, `product.attribute`	Base plans with recurring frequency attributes and add-on variant pricing (+560).
Quotations & Subscriptions	Odoo Subscriptions	`sale.order`, `sale.subscription`	Contract state transitions: Draft → Quotation → Confirmed → Active → Closed.
Invoices & Automated Tax	Odoo Invoicing / Accounting	`account.move`, `account.tax`	Automated recurring invoice generation, standard GST/VAT tax calculation.
Counter Payments	Odoo Point of Sale / Bank	`account.payment`	Multi-method reconciliation (Cash, UPI, Card, Bank) settled against invoices.
Front-Desk Check-in	Odoo HR & Attendance	`hr.attendance`	Fast daily attendance tracking, timestamped logs, and access validation.
Renewal & Expiry Alerts	Marketing Automation / Actions	`base.automation`, `mail.template`	Automated 7-day advance triggers, automated invoice drafts, follow-up logs.

3. Role-Based Access Control (RBAC) & Security Policy

Per the hackathon security guidelines, access is strictly partitioned across three roles with enforced password standards.

🔒 Security & Password Compliance

- **Strict Email Uniqueness:** Exactly one account per verified email.
- **Enforced Complexity:** Minimum 8 characters, at least 1 uppercase letter, 1 lowercase letter, and 1 special character (`[!@#$%^&*]`).
- **Session Security:** Role-aware route guard prevents unauthorized API access.
- **Audit Logging:** All payments and check-ins record the executing staff ID.

👥 Role Permissions Matrix

- **Gym Admin:** Complete system control, plan creation, variant pricing, discount rules, tax rates, staff user management, and executive financial reports.
- **Front-Desk Staff:** Member registration, quotation drafting, invoice confirmation, payment recording, attendance check-in, and renewal triggers.
- **Member Portal:** Self-service view of active plan, expiration date, payment history, attendance logs, and renewal requests.

4. Core Business State Machines

A. Subscription Lifecycle State Machine



B. Invoice & Payment Settlement State Machine



5. Relational Data Model & Field Specifications

FitFlow stores business data in normalized entities that support rapid query speeds (<2s) even with thousands of active member records.

Entity	Key Fields & Data Types	Business Rules & Validation Logic
User / Member `res_partner`	`id (UUID), email (UK), password_hash, full_name, phone, role (admin|staff|member), status, created_at`	Email must be strictly unique. Password validated for 8+ chars, upper, lower, and special character.
Membership Plan `product_template`	`id, name, billing_interval (daily|weekly|monthly|yearly), interval_count, base_price, cost_price, min_qty, available_from, available_to, allow_pause`	Governs contract duration. Defines whether a plan can be paused (e.g. for injury) or auto-closes on expiry.
Plan Variant `product_attribute`	`id, plan_id (FK), variant_name (Trainer|Locker|Diet), additional_price, is_active`	Adds specific line addons to base subscription (e.g., Personal Trainer adds ₹500 / $50).
Discount Rule `discount_rule`	`id, code (UK), discount_type (percent|fixed), discount_value, min_purchase, max_uses, current_uses, valid_from, valid_to`	Admin configured. Automatically validates cart thresholds and decrements remaining uses.
Tax Rule `account_tax`	`id, name (e.g. GST 18%), rate_percentage (18.0), is_active`	Automatically applies tax on taxable base lines before final invoice confirmation.
Subscription `sale_subscription`	`id, sub_number (UK), user_id (FK), plan_id (FK), state, start_date, expiry_date, subtotal, discount, tax, grand_total`	Moves through state machine. Auto-calculates next billing date based on plan interval.
Invoice `account_move`	`id, inv_number (UK), subscription_id (FK), user_id (FK), state (draft|confirmed|paid|overdue), due_date, total, balance_due`	Created automatically on subscription confirmation or renewal. Prevents double-billing.
Payment `account_payment`	`id, payment_number (UK), invoice_id (FK), amount, method (cash|card|upi|bank), payment_date, reference_note`	Reduces invoice `balance_due`. Changes invoice state to Paid when balance reaches 0.
Attendance Log `hr_attendance`	`id, user_id (FK), subscription_id (FK), check_in_time, status (granted|warning|rejected), notes`	Instant desk verification. Rejects entry if subscription state is not Active or invoice is overdue.

6. Responsive UI Architecture & Operations Ergonomics

🖥️ Desktop Command Center (1280px+)

- **Navigation:** Persistent left sidebar with real-time badges (e.g., "Renewals Due: 5").
- **Quick Search:** Global `Ctrl + K` shortcut to search members instantly by phone or name.
- **Split Workflows:** Two-column forms with live calculations (tax & discount totals update dynamically).
- **Side Drawer:** Full member 360° profile view slides in without leaving the current table.

📱 Mobile Front-Desk View (<768px)

- **Bottom Bar:** 4 high-frequency touch targets (Dashboard, Check-in, Members, Invoices).
- **One-Thumb Check-in:** Front desk staff can search a member and tap large green "CHECK IN" button with one hand.
- **Card Stacks:** Wide desktop tables convert into touch cards with color-coded status pills.
- **Haptic / Visual Feedback:** High contrast confirmation banners for check-ins and payment receipts.

7. Step-by-Step Implementation Plan ("Develop One-by-One")

To ensure a rock-solid, fully testable submission for Odoo Make-A-Ton, development is sequenced into six self-contained milestones:

Milestone	Core Deliverables	Key Implementation Steps	Acceptance Criteria
Phase 1 Foundations & Auth	RBAC Security & Data Models	• Setup standalone responsive framework. • Implement User model with password rule verification (8+ chars, upper, lower, special). • Seed demo dataset (Admin, Staff, 5 active/expiring members).	Login works with role switching (Admin, Staff, Member). Passwords strictly enforced.
Phase 2 Plans, Variants & Quotes	Product Catalog & Quotation Engine	• Plan configuration (monthly, quarterly, yearly). • Variant builder (Personal Trainer +₹500, Locker +₹200). • Discount rules engine (percentage & fixed limits). • Quotation builder moving from Draft → Confirmed.	Quotes calculate dynamic totals with variants and discounts automatically.
Phase 3 Billing & Payments	Automated Invoicing & Tax Calculation	• Auto-generate invoice from confirmed quotes. • Apply 18% tax calculation automatically. • Record payments across Cash, UPI, Card. • Transition invoice from Confirmed → Paid.	Invoice balance drops to zero; payments audit log recorded; no duplicate billing.
Phase 4 Desk Operations	Quick Check-in & Attendance Logs	• Instant member search by name or phone. • Real-time membership check: Active → Granted; Expired → Alert prompt. • Attendance history timeline. • Inactivity detector (>14 days absent).	1-tap check-in on mobile; clear visual alert if member membership has lapsed.
Phase 5 Renewal Automation	7-Day Expiry & 1-Click Renewals	• Dynamic filter: memberships expiring in ≤7 days. • 1-Click Renewal: extends dates and generates renewal invoice. • Member Portal view: member logs in to view membership pass and invoice receipts.	Renewal seamlessly extends subscription without creating orphaned records.
Phase 6 Executive Dashboard	Finance Lead Analytics (<2s) & Odoo Export	• Real-time KPI cards: MRR, Active Count, Overdue Invoices, Today's Check-ins. • Filterable tables with instant search (<2s response). • Odoo Integration API Payload export (JSON mapping to `sale.subscription`).	Fast, reactive dashboard providing complete visibility into gym recurring revenue.

8. Hackathon Presentation Checklist

📋 The Judging Criteria Checklist

- ✅ **Subscription Depth:** Multiple frequencies, variants, discounts, quotation templates, and automatic renewals covered.
- ✅ **Odoo Philosophy:** Connected app architecture (Sales + Accounting + Attendance + CRM) without siloed data.
- ✅ **Operational Usability:** High-touch front-desk workflows (attendance, instant alerts) tested for speed.
- ✅ **Security & RBAC:** Password complexity and multi-user portal segregation strictly implemented.

💡 Pro-Tips for the Pitch

- **Show, Don't Tell:** Walk through the persona of member "Alex" from initial quotation to desk check-in.
- **Highlight the "Billing Day" Button:** Demonstrate batch recurring billing running in seconds.
- **Demo on Mobile:** Open the mobile view on a phone or DevTools to prove true front-desk ergonomics.
- **Emphasize MRR:** Show the Finance Dashboard calculating Monthly Recurring Revenue instantly.